

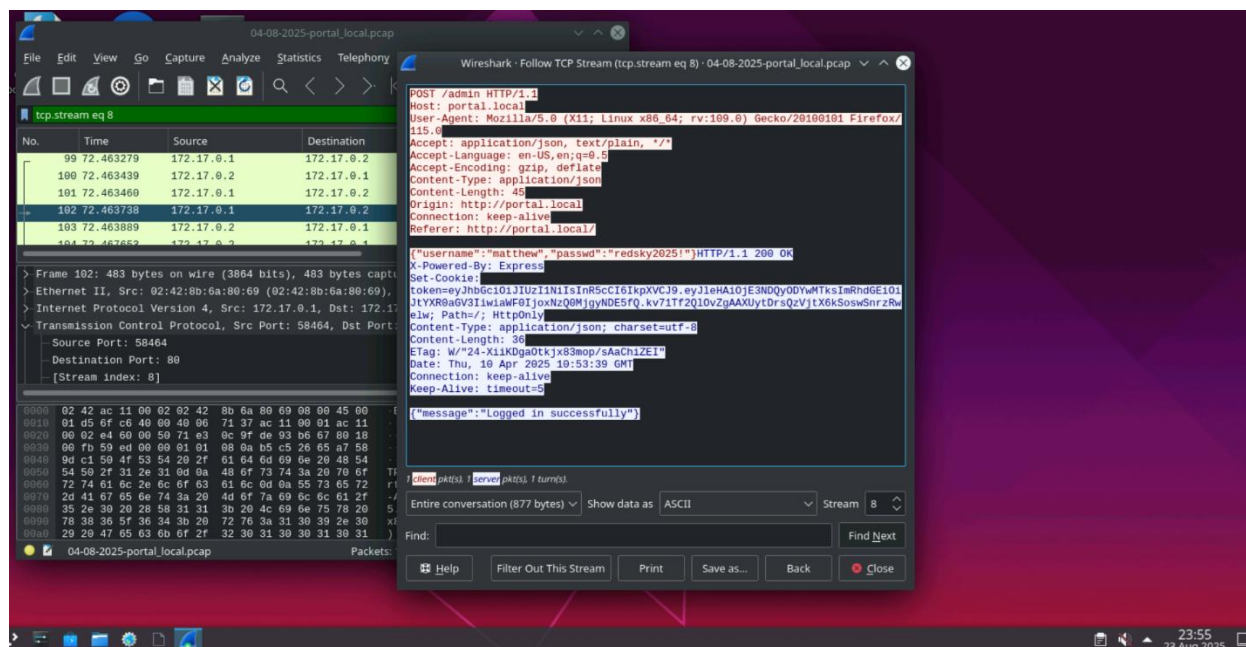
CASE: Possible Brute force attack

8/23/2025

Looking at unsuccessful login attempts I became concerned of a brute force attack. Upon further inspection I noticed that important information was being transferred over HTTP versus the secure HTTPS. I launched Wireshark and clicked open, I then opened the packet capture file (04-08-2025-portal_local.pcap. I then began to analyze failed and successful login attempts using HTTP in the filter section to find the required website, portal.local.

After finding the website I put it in the chromium browser and met with the username and password internal website. I then went back into Wireshark and searched through packets and ran the follow command on a few packets and chose to follow the TCP stream. It was there I found the username Matthew and the password transmitted over plain text. Upon testing the username and password I was able to brute force our system and get into confidential information. After recording the breach protocols were changed to HTTPS to ensure private data encryption.

Here is where I found the required data, username, password and website.



Terms defined

HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
Packet	A form of data send from one system to another
TCP	Transmission control protocol

Finding	Finding details	Description
Username	Matthew	Username found in data stream
Password	redsky2025!	Password found in plain text
Website	Portal.local	Internal network website